In-plant Milk Run Decision Problems

Mohammed Alnahhal, Asep Ridwan, Bernd Noche
Transport systems and logistics
Duisburg-Essen University
Duisburg, Germany

***Abstract*— This study investigates the general framework of the decisions in in-plant milk run systems, and also the other related decisions that affect or affected by milk run systems. Emphasis was placed on lean-oriented decisions and objectives. This was accomplished conceptually using extensive literature review. The study shows that there is a lack of studies regarding in-plant milk run system as a whole, and also in certain areas of the system such as manual feeding. The study shows also a tendency in some studies to deviate from lean manufacturing philosophy looking for optimality based on restrictive objectives. Furthermore, some differences in literature in defining some concepts related to in-plant milk run system were illustrated. The study presents new directions for future research especially those related to fast-response delivery in very changing environments.**

***Keywords*— In-plant milk run; decision making, lean manufacturing, JIT**

I. Introduction

In manufacturing facilities, there are two types of activities, namely, making goods and operations needed to deliver these goods. The second type is “logistics”. If it is inside the boundaries of the facility, containing the material flows between plants and between production lines within a plant, then it is under control of the management of the company. However inbound and outbound logistics are outside the control of this management [1].

Production logistics is one of the most important aspects that must be optimized to enhance the performance of assembly systems. Automotive companies started to utilize the concept of in-plant milk run, in which a train or manual cart takes bins full with parts from inventory position (a store) to consumption points (stations) in a predetermined route and schedule [2]. So, a group of stations is supplied by the same train in the same route and in the same service period. Bozer and Ciemnoczolowski [3] described milk run systems as “route-based, cyclic material handling systems that are used widely to enable frequent and consistent deliveries of containerized parts on an as-needed basis from a central storage area (the ‘supermarket’) to multiple line-side deposit points on the factory floor”. Fig 1 shows the components of milk run system. This system depends on frequent delivery of small amount of materials. Usually, there are two separate tugger routes — one delivering purchased parts, and the other removing finished goods [4]. In spite of its importance, the principle of in-plant milk run did not take the appropriate attention by researchers [5].

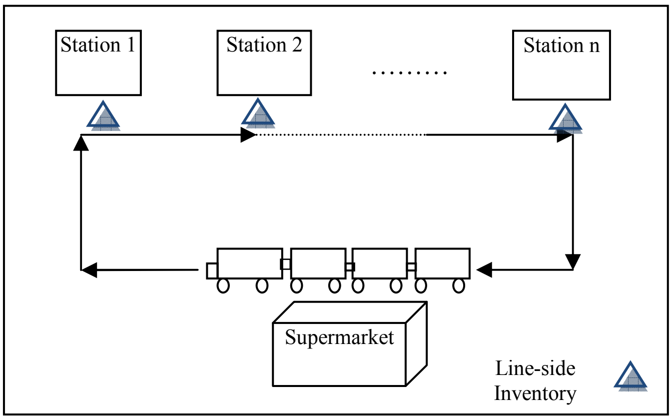

Fig.1. In-plant milk run system

Forklift is not appropriate for repeatable flow of items between the same locations [1]. In this system, large quantities of purchased parts arrive at the receiving dock, typically on pallets or in boxes, and are delivered directly to the production floor in an unorganized manner. So it is difficult in such a system to know how many purchased parts are on the floor, cells running out of parts, and operators unable to sustain takt time because they have to go searching for parts [6]. Using frequent delivery cycles results in more space efficient, transparent and less variable process. Conventional material feeding transporters such as forklifts or automated guided vehicles (AGVs) are inappropriate to address the special demands of a lean production system [2].

Pushcarts or trains of tow carts picking up and dropping off quantities of materials that are usually less than full pallet loads, provide a cheaper, more reliable, and more predictable service [1]. Material supply by this system occurs on a just-in-time basis [7]. Moreover, tuggers or tow-train have been proven to reduce material handling distances within facilities [8]. Furthermore, tuggers are safer, more maneuverable, and less expensive than forklifts [4].

Because milk run system provides material in small lot sizes with high frequency, it is used especially in automotive industry in which the number of unique assembled products on the production line is constantly increasing. Actually, in the automotive industry, a large number of different materials should be delivered to production line. The space for material provision, which is near the assembly line, is not adequate. There are only few bins that can be stored next to the assembly line. As a result, the company has to find a rapid, continual and reliable in-plant supply process to deliver different materials. The in-plant milk run system can satisfy the requirement above. That is the reason, why in-plant milk run system is applied in automotive industry. However, because of its complexity in planning and dimensioning, in-plant milk run system can hardly be applied in other industries [2].

In this study, the different decisions problems in in-plant milk run systems will be investigated depending on extensive literature review. The remainder of this paper is organized as follows. Section 2 is about literature review of using in-plant milk run and, in particular, the conceptual previous studies about decisions in milk run systems. Section 3 will focus on summarizing decisions and objectives found in the literature regarding milk run systems. Section 4 considers the other related decisions affecting or affected by the milk run system. Section 5 will shed the light on lean-oriented decisions and objective in milk run systems. Section 7 presents directions for future research. And finally, section 6 will contain conclusion.

II. LITERATURE REVIEW

Inbound logistics means transporting parts from suppliers to the manufacturing plant, but outbound logistics means transporting parts from the manufacturing plant to its customers. As stated before, most of the literature about milk run systems investigated inbound or outbound logistics. Maybe one reason for that is that external milk runs are more complicated to set up than in-plant milk runs [1]. Brar and Saini [9] reviewed the literature on milk run logistics outside the manufacturing plant especially in automotive industry. Most of the literature found about the internal milk run systems was about feeding assembly lines. For example, Alnahhal and Noche [10], Faccio et al. [11], and Faccio et al. [12] studied the use of milk run system to feed mixed model assembly lines. However, milk run operator can also move the items to the regular manufacturing stations and from the assembly lines to the testing, repairing, final packaging, and output buffer [1]. Using in-plant milk run system was investigated for work improvement by several studies such as Sihn and Schmitz [13], Wong and Wong [14], and Kasperk et al. [15]. Furthermore, milk runs and their tugger trains were recommended by many studies to make improvement inside facilities [16, 17, 18, 19].

In-plant milk run system supports lean manufacturing philosophy in which seven wastes are reduced [20, 21]. These wastes are

- Waste from producing defects
- Waste in transportation
- Waste from inventory
- Waste from overproduction
- Waste of waiting time
- Waste in processing
- Waste of motion

The second waste is the focus of our study. If material is delivered just in time, another type of waste, namely, inventory cost, is decreased. Also, providing materials, after they are needed, causes starvation for workstation. This will cost a lot because of lost sales and resources underutilization [11]. So in JIT, material must be supplied not before and not after it is needed. JIT is a major factor in Toyota production system in an assembly industry such as automotive manufacturing. Only the necessary parts, at the necessary time, in necessary quantity are delivered, and in addition, the stock on hand is held down to a minimum. JIT is able to shorten the lead time from the entry of materials to the completion of job. It also discloses existence of surplus equipment and workers [22]. To achieve lean manufacturing, some tools that form the “house of lean” are used. These tools contain value stream mapping, standardized work, 5S, and so on. Later we will explain their relation to milk run system.

A. Advantages and Disadvantages

The most important decision is to use milk run or not, based on its advantages and disadvantages, which depend on the particular conditions of each environment. The positive impact of using in-plant milk run system was investigated in some studies. Hanson and Finnsgard [23] presented a case study in which the transition from using mainly the forklifts to milk run tugger trains to replace the pallet loading units with plastic containers loading units. This was to achieve compact assembly stations with parts presentation that could support efficient assembly. As a result of the changes, it was possible for the company to reduce the work force at the assembly line by 13.5 FTE (full-time equivalent) assemblers. Another result was reducing the inventory size beside the workstations. Moreover, they get rid of the capacity problems, which in turn result in unpredictable delivery times and delayed deliveries. Furthermore, tuggers were more directly connected to consumption than the forklift system. Moreover, Rother [24] showed several advantages for using in-plant milk run tow trains after using them in an automotive company in Germany such as increasing productivity, enormous cost savings, 15 % less empty runs, reducing traffic volume, less amount of resources, saving a huge amount of space, and enhancing safety.

Moreover, Finnsgard et al. [25] examined in a case study how the choice of materials exposure impacts workstation performance, in terms of non-value-adding work, space requirements and ergonomics. The transition from pallet sized unit load transported by forklifts to smaller plastic containers sized unit loads transported by tugger train has a great impact where the space required for materials was reduced by 67%, non-value-adding work was decreased by 20%, and walking distance was reduced by 52%. Furthermore, the ergonomics for the assembly operator improved greatly, with a 92% reduction of potentially harmful picking activities, thereby almost eliminating potentially harmful body movements. The positive ergonomic effect of using narrow bins compared to using big boxes was investigated in a study by Neumann and Medbo [26] which also considers the effect on performance times, walking distances, and layout space requirements.

However, the planning process for transition from traditional system to milk run must take a lot of deep analysis and thinking, because sometimes the costs for this transition are very high. For example, Hanson [27] presented a case study about “Minomi” which is a unit load where no container is used, and identified the effects of using minomi in the materials supply within an assembly plant. To facilitate using this concept, milk run tugger train was used instead of the traditional forklift system. However, in that case study the parts should, also after the introduction of minomi, be placed and handled in their original steel containers in order for efficient handling and storage to be maintained. With this solution, forklifts were still required for lifting the containers from the storage area onto the tugger train, meaning that the total number of handling operations increased in the materials supply operations. The combination of increased handling necessary for each delivery and increased delivery frequency resulted in a 326 percent increase in the man hour consumption for performing the deliveries. Furthermore, in the case study illustrated above in the study by Hanson and Finnsgard [23], the main disadvantage of the new system was in introducing a new activity which is repacking the pallets coming from the suppliers to the new plastic container size and also the sequenced deliveries which needed additional 13 workers. So applying the principle of in-plant milk run is not always without side effects. As a matter of fact, beside additional work on transferring materials from big containers to smaller bins, milk run systems increase the number of bins in the system and increase the stress on routing operators [4].

B. *Designing Milk Run Systems*

The planning of designing milk run systems was investigated in literature. Abele and Brungs [28] formulated fundamental concepts to optimally design and develop milk runs. In a study by Alizon et al. [29], milk run trains and forklift system were used in the same time to feed the same assembly workshop. Michael and Claudia [30] presented several characteristics of milk run and showed that a higher delivery frequency has a positive effect on a marked reduction of inventory volume at the production stations. Furthermore, they presented approaches regarding demand requirements, supply, and transport control that must be considered in the system so that efficient and ecologically beneficial delivery system can be implemented.

Classification of milk run design decisions was also investigated in previous studies, but in a limit range. For example, Droste and Deuse [2] defined four design alternatives the decision maker should focus on when he wants to use milk run system, these alternatives are material control, warehouse, means of transport, and Point of Use (PoU). In material control, the planner can decide between just in time, just in sequence, and also between Kanban and CONWIP systems. In the warehouse, there is direct and parallel picking from flow rack, pallet rack, shelf, etc. In means of transport, the planner can consider the types of conveyor, manually operated cart or propelled towing train. In PoU, the planner can consider provision directly to stations in flow rack or provision on a spot close to station. However, Droste and Deuse [2] did not concentrate on other importance decisions such as routing and scheduling.

Moreover and based on the observations in real manufacturing environment and limited literature, milk-run distribution problem was categorized and explained in a study by Kilic et al. [5]. Their classification was related to the routes and their time periods. There are three categories of problems: general assignment problem, dedicated assignment problem, and determined time periods assignment problem. In “general assignment problem”, the routes and time periods are not known. “Dedicated assignment problem” is the second category where the routes are known but the problem is to determine the time periods of the vehicles on the related routes. The last category is “determined time periods assignment problem” in which time periods are known but the routes are not known. As obvious in the study by Kilic et al. [5], the concentration is on routes and time periods, which means that they did not consider the other types of decisions.

As can be seen from the previous literature, there is a need for a comprehensive study that not only takes into consideration the decisions problems in in-plant milk run system, but also the other related decisions that affect or affected by the in-plant milk run system decisions. This is what this study tries to do.

III. IN-PLANT MILK RUN DECISIONS AND OBJECTIVES

In-plant milk run as a principle is new. Its complex nature from different sides is not well explained in literature. One of the objectives of this study is to give the reader the knowledge about the environment of milk run application and its decisions since it needs a lot of complex planning tasks. The study also explains the relations among different decisions, and warns the decision makers not to make fast judgment about in-plant milk run before taking enough planning into consideration. It also gives the researchers hints about future research trends regarding in-plant milk run especially the areas that have not been covered yet by literature.

A lean material-handling system should deliver materials to the operators' fingertips. This step involves determining delivery aisles, selecting a conveyance method, such as tuggers, determining the stops and delivery points for the route, and creating correctly sized point-of-use gravity racks at delivery points [6]. Drivers should fill the racks from the outside, so they don’t interrupt the working operators [4].

Harris and Harris [6] defined 5 major steps for the transition from traditional material flow to the lean material flow using the milk run systems:

1. Develop a plan for every part (PFEP).
2. Build the purchased-parts market
3. Design delivery routes
4. Implement pull signals
5. Continuously improve the system

During doing these five steps, most of the decisions can be made. For more information about applying these steps, the reader may refer to Marchwinski [4] who presented a case study about that. Enough information should be available for every part in the system in the first step. In the second step, the major steps in designing the market are figuring the maximum inventory level for each part, and establishing rules for operating the market. After determining the maximum inventory level, the maximum space level for the inventory should be determined. In the third step, decisions such as choosing the transporter type, routing, scheduling, design of racks, and number of containers in the racks can be made. In the fourth step, the configuration of the kanban system can be determined such as the number of kanbans based on the type of routes, coupled or decoupled (will be explained later). Different standardized procedures should be followed to continuously improve the work.

Table 1 shows different studies with the decisions investigated, objectives, type of transporter (tow train (T) or manual cart (C)), and the feeding system used (pull Kanban (K) or demand-oriented (D-O)). D-O means working based on certain and known stations demand for just short periods. As can be seen from the table, there are some studies which used demand-oriented system in milk run. These studies investigated assembly lines in automotive industry. Difficulties come in the case of high variant mixed model assembly when the demand of stations changes every day because of the ever changing daily production sequences. What makes it even harder is the fact that materials must be there when they are needed to avoid disruptions in the assembly process. So in this case, demand-oriented system performs superior to the popular Kanban systems [31]. In this demand-oriented system, using small frequent deliveries of raw materials facilitates using the JIT principle, and "forecasting" the demand of stations is only for three or four days. This could be better than waiting to be ''surprised'' by low stocks, which may make it very necessary to make necessitate emergency deliveries if the next scheduled stopover at the station is too far off [32]. As obvious in table 1, most of the researches concentrated on the case of using tow trains. The only decision about manual carts was just to determine to choose them or to choose the tow trains instead. The decisions can be summarized as follows:

- Choosing kind of conveyor (KC)
- Choosing demand-oriented or Kanban systems
- The number of kanban required in the system (K)
- Determining number of trains (N)
- Number of handling operators (NO)
- Routing problem (R)
- Scheduling problem (S)
- Determining service period (P)
- Loading problem (L)
- Frequency of routing (F)
- Parts service level (SL)

Decisions introduced by Droste and Deuse [2] which were mentioned in the previous section can be added to the list. In table 1, the study by Battini et al. [33] is a conceptual one reviewing the previous researches investigating the decision problems in the case of using decentralized supermarket system. Choosing the means of transport depends mainly on the distance traveled beside the layout of the facility. Generally, for long distances, tow trains are better. On the other hand, sometimes there is no enough space for the tow trains. As stated by Baudin [1], "push carts are not adequate to deliver thousands of items in box quantities at multiple locations every half hour, which is commonly needed in the automobile industry". The number of trains is a decision variable and an objective in the same time where the objective of course is to reduce it. As stated by Kilic et al. [5], the route for the train can be fixed or variable.

TABLE I. IN-PLANT MILK RUN STUDIES: DECISIONS AND OBJECTIVES.

Study	K	D-O	T	C	Decisions	Objectives
Alnahhal and Noche [10]		X	X		R, S, L	IC, SV, N
Álvarez et al. [35]	X			X	KC, F, R	IC, U
Battini et al. [33]		X	X		R, S, L	IC
Ciemnocz-olowski and Bozer [36]	X		X		K	SP
Costa et al. [37]	X		X		N	N
Domingo et al. [34]	X			X	KC, F, R	IC, U
Emde and Boysen [38]		X	X		R, S	IC
Emde et al. [39]		X	X		L	IC
Faccio et al. [11]	X		X		N, K	U
Faccio et al. [12]	X		X		K, NO, SL	IC
Golz et al. [31]		X	X		R, S	N
Kilic et. al. [5]	X		X		R, P	N, D
Satoglu and Sahin [7].	X		X		R, P	MH, IC

In the variable one, the train can have different routes. Domingo et al. [34], for example, defined different routings for different hours in the day, since it is not necessary to visit each job with a high frequency. However and from the lean manufacturing point of view, the fixed routing is better since it standardizes the work. The periods in scheduling can also be fixed or variable.

Emde and Boysen [38] showed that the variable periods are optimal for reducing inventory costs. However and from the point of view of lean manufacturing, it is better to standardize the system by stabilizing the periods. Some studies even considered fixed periods as a must to call the system as milk run such as Bozer and Ciemnoczolowski [3]. Moreover, Hanson and Finnsgard [23] presented a real case study in which the time period of the tugger train was constant. The routing and scheduling are related to each other. Some studies assumed that routing is input and they investigated the scheduling problem. On the other hand, some studies assumed that the periods are input, and they investigated the routing problem. However, other studies such as Emde and Boysen [38] studied the routing and scheduling together.

In the case that for some periods of time the demand is more than the capacity of the trains, loading problem, in which some parts are loaded in previous non bottleneck periods, will be necessary. Therefore, the decision here is about determining for each route which parts and by which quantity should be loaded on the train to minimize the total inventory. This was in the demand-oriented system formulated by Emde et al. [39]. However in the pull system, there is another decision problem regarding kanbans which control the precise times and quantities of parts delivered to stations. It is important to precisely control the number of kanban circulating in the system. The first step is to determine how frequently to deliver material to stations, and whether the route is "coupled" or "decoupled." In a coupled route, the tugger driver loads carts in the market and drives them to the stations, and delivers them to the point of use. In a decoupled route, the work is divided between a market attendant who loads parts, and the driver who delivers them. Decoupled routes require two sets of carts but they improve labor utilization, so routes can be longer and have more carts [6]. In decentralized supermarket system, the decoupled routes seem not very effective because in this case a market attendant should be there for every supermarket which increases the labor costs. Ciemnoczolowski and Bozer [36] determined the number of kanbans and considered the effects of the number of kanban, the physical capacity of the tugger, and the cycle time on workstation starvation defined as the fraction of available time a workstation is idle due to lack of parts.

The objectives contain minimizing inventory costs (IC), system variability (SV), number of trains (N), starvation probability (SP), distance traversed (D) and total material handling costs (MH); and increasing utilization of workers (U). Beside the decisions and objective, a very important aspect about in-plant milk run system is its constraints. There are restrictive dimensions of in-plant milk run systems which are time, capacity and ergonomics and have to be considered properly in order to achieve an efficient in-plant milk run system [2]. Obviously, the necessary handling and transportation time of a milk run has to be less than or equal to the milk run cycle time in order to maintain the planned material provision takt. The capacity of the tow train must be considered. The physical strain on the material handlers must also be considered. Beside these restrictions, different restrictions were found in the literature such as starvation introduced by Bozer and Ciemnoczolowski [3] and Ciemnoczolowski and Bozer [36] where they took into consideration determining the level of workstations starvation for given system parameters. Other studies such as Emde and Boysen [38] assumed in their model that the number of part shortages must be zero.

IV. OTHER RELATED DECISIONS

In-plant milk run system does not stand alone. There are many decisions that are related to this system. Some of them can be considered as inputs to the milk run system and others can be considered as outputs. The in-plant milk run was considered in a study by Kovács [40] as input for the storage assignment problem which involves the placement of a set of items in a warehouse in such a way that some performance measures are optimal. In that case, the most important input from the milk run is routing. In another study by Alizon et al. [29], in which the assembly work shop was fed by tow trains and forklifts in the same time, the assignment problem was performed. However, the major concern of that study was the delivery of components performed by forklifts, so it was assumed that the routes followed by trains are fixed. The problem is then to optimize forklift delivery loops by including congestion caused by loading/unloading in a potential blocking aisle.

On the other hand, according to Ramesh [41], a good layout can lead to an effective implementation of milk run concept for the smooth flow of in-process material. Another important input decision that can be investigated is storage centralization/decentralization decision. For example, Battini et al. [42] considered investigating this decision where they studied the usage of supermarkets which are “decentralized storage areas scattered throughout the shop floor which serve as an intermediate store for parts required by nearby line segments”. Some of the advantages of decentralized supermarket system are “shorter delivery times by being closer to the consumer (i.e., the assembly line), freight consolidation by being supplied by industrial trucks, and faster turnover by stocking and delivering parts just as needed”. “On the downside, supermarkets consume space on the factory floor, which is scant and expensive” [32]. Furthermore, a cost model was presented in a study by Sargent et al. [43] where the model can be used to determine if further consideration should be given to decentralized storage in a facility currently utilizing centralized storage. The cost model consists of examining the trade-off between the savings in material handling flow costs due to moving from centralized to decentralized storage and the additional costs associated with implementing and utilizing decentralized storage for a designated period of time. Satogluet al. [44] evaluated the conversion from central storage to decentralized storages in cellular manufacturing environments using activity-based costing. Furthermore, Alizon et al. [29] studied the flow of materials from several internal warehouses to supply the same assembly workshop. Moreover, in a study by Hanson and Finnsgard [23], a case study was explained in which the material feeding was organized by using three drop zones which were close to three assembly lines. These drop zones get the material from the AS/RS system. Then material was transported from the drop zones to the different stations in the assembly lines. One the other hand, in a study by Marchwinski [4], a case study was presented about a company which had 4 storage areas feeding the assembly lines using the traditional forklift system. Then when the company wanted to use milk run tugger train, it used one central supermarket and considered that as an advantage to reduce the inventory costs. Besides, its location near the receiving dock permits quick delivery from receiving to storage racks.

So if the decentralization decision has been made, another decision also must be made which is the location of the decentralized supermarkets which was investigated by Emde and Boysen [32]. This means that at first, a decision has to be made about using centralized or decentralized supermarkets. If the decision is to use the decentralized supermarket, the location problem is a must. After that, other milk run decisions can be studied. For example, Emde and Boysen [38] studied the scheduling and routing problem of the tow trains, and Emde et al. [39] studied the loading problem. However, if the decision is to use the centralized system, there is no need for the supermarket location problem such as the way in the study by Golz et al. [31]. The general framework of these decisions can be shown in Fig. 2.

It is good to mention here the definition of "supermarket". There is no universal meaning for it. Some studies considered supermarket as a central storage area such as Golz et al. [31], Bozer and Ciemnoczolowski [3], and Ciemnoczolowski and Bozer [36]. Other studies considered it as decentralized storage area such as Battini et al. [42], Emde et al. [39], Emde and Boysen [38], and Emde and Boysen [32]. Other studies define the supermarket as central picking store in which the kitting process is performed [4]. All these types are also different from the famous "Kanban Supermarket" used in lean Manufacturing [44, 45]. On the other hand, Marchwinski [4] and Harris and Harris [6] named the central warehouse holding the purchased parts as "market". There are also other decisions that interact with the decisions in the milk run systems such as line stocking and kitting which are two alternative material supply systems that are common in assembly systems. Line-stocking systems supply components to the assembly line in homogeneous individual component containers stored close to the assembly workstations at the border of the line. On the other hand, kitting systems combine various components into one heterogeneous package according to a future assembly schedule. The required quantity of components is stored in kit containers at the border of the line [45].

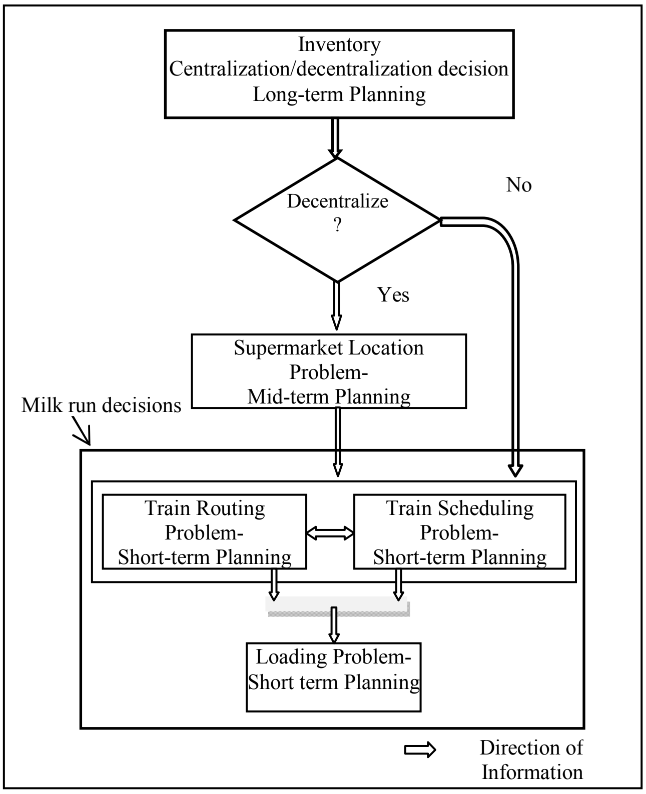

Fig. 2. General Framework of the milk run system found in the literature in the case of demand-oriented system.

The kit assembly operation can be performed in a central picking store or in decentralized areas close to the assembly stations. In the case of using central picking store, the parts are transported to this store from the central warehouse using the tow train or the forklifts. So besides its working in the milk run tours, the tow train can also be used for handling material from the central warehouse to the central picking store of kitting. In this case, the milk run route feeding different stations does not start from the central warehouse but from central picking store. Fig. 3 summarizes the input and output decisions related to milk run system found in the literature.

V. LEAN ASPECTS IN MILK RUN SYSTEM

The advantages previously mentioned such as reducing inventory costs, workforce requirements, the space required for materials, non-value-adding work, and walking distance coincide with the principle of lean which is about minimizing wastes. Moreover, introducing milk run help in organizing the shop floor, and this coincides with the 5S principle. The basic principle of in-plant milk run is providing small and repetitive amounts of materials for the workstations. This is done according to the JIT principle to minimize inventory costs which coincides with the lean manufacturing philosophy. In most of the cases, the pull system using kanbans was used in the milk run systems. However, in some cases as stated before, the demand-oriented system was also used to predict the demand only for few days which does not contradict severely with the JIT principle. Another tool of lean manufacturing is value stream mapping [44]. External milk runs were extensively used in the literature in value stream mapping. However, only few studies considered using in-plant milk runs in this tool such as the study by Álvarez et al. [35] which used in-plant milk run inside the value stream mapping for redesigning an assembly line. According to the same study, using kanban alone without internal milk run is not enough for reducing the total inventory. One of the most important tools of lean manufacturing is standardized work. The effect of this tool is very obvious in stabilizing the routes periods and the routes themselves. The periods of trains' routes should be the same even if this is not optimal, because changing the periods for every route is very confusing for material handlers. The same is also for standardizing the trains' routes. This standardization in routing and scheduling was emphasized by several studies such as Abele and Brungs [28].

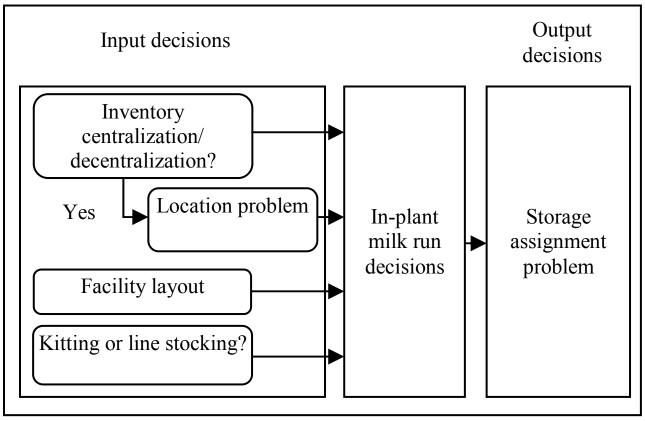

Fig. 3. Input and output decisions, found in the literature, related to milk run

Some studies such as Emde et al. [39], Emde and Boysen [38], Emde and Boysen [32] assumed using “shooter racks” in tow train systems. These special gravity flow racks are used to automatically load bins on the wagons using elastic springs. These racks reduce the length of a stopover to a few seconds. However, not all the automotive companies use this system. From the point of view of lean manufacturing which makes more emphasis on managerial (not technological) enhancements, consideration should be more for manual loading and unloading.

VI. DIRECTIONS FOR FUTURE RESEARCH

By looking at the objectives found in the literate in table 1, we realize that these objectives mainly concentrate on reducing the costs of the system. However, there is lack of studies regarding fast response of the system to any changes in material demand. Failure of the system to this fast response means increasing starvation of workstations. One direction is to investigate the impact of changing the type of route from coupled to decoupled one on the response speed of the system. Another factor is the impact of using the centralized or decentralized supermarket system. Previous studies concentrated on analyzing the decision mainly based on the system costs. Future research, however, can investigate the optimal degree of centralization/ decentralizing of inventory based on the speed of the system. Having decentralized supermarkets will make the inventory not far away from the workstations, and this means that if a shortage in material takes place, the replenishment of material will not take too much time. Another factor that enhances the response speed of the system is using technologies such as radio frequency identification (RFID) which can be used to send the signals for material replenishment. Its effects have a good research opportunity in the future. Using RFID and decoupled routes together gives the opportunity to decrease the service period and decrease the lead time. Besides that, determining the size of resources such as number of tow trains should be reinvestigated based on the new objective. Decisions such as route length, frequency of routing, number of containers per route, and container size can also be investigated to enhance the response speed of the system. Keeping spare trains' capacity contributes very well in this direction. Beside these operational decisions, fast response-delivery can be considered in higher levels such as supplier selection. One criterion may focus on the agreement with suppliers to supply materials in final small container size to eliminate repackaging time.

Future studies can also focus on combining the decisions of the physical characteristics of the tow trains and the managerial aspects of using it. For example, usually there are several types of tow trains available with different widths, and using the appropriate type depends on the aisles width which must be larger than the train width plus a sufficient clearance. Therefore, choosing the appropriate type of tow trains depends on the layout of the factory. It may be wise to consider the two decisions together in the same time. Another aspect is the turning radius of the train which also depends on the layout. This radius also depends on the tow train length which depends on the number of trailers attached to the tow train. Using long tow trains saves material handling costs; however, if the number of the trailers is too large, it will affect the safety, and sometimes could not be attainable because of the layout of the factory. Attention should also be made for the even distribution of the weight over the trailers of the train. Another dimension that can be considered in the future research is the congestion of the tow trains inside the facility during the loading and unloading of these trains in narrow aisles. This problem can be minimized by reducing the number of loading/unloading stops of the tow train without increasing starvation probability or inventory costs.

VII. CONCLUSION

This conceptual study focused on investigating the general framework of decisions usually made regarding in-plant milk run systems using extensive literature review. The usual addressed objectives were illustrated. The other external decisions which can be considered as inputs for or outputs from the milk run decisions were investigated. The decision maker should take into consideration the different aspects of lean manufacturing in the design of the milk run. Future studies can concentrate on investigating these decisions especially those which have not covered sufficiently in literature to enhance the speed of the response of the system to any changes in parts demand or capacity shortages.

REFERENCES

[1] M. Baudin, Lean Logistics: The Nuts and Bolts of Delivering Materials and Goods. New York: Productivity Press, 2004.

[2] M. Droste and J. Deuse, “A Planning Approach for In-plant Milk Run Processes to Optimize Material Provision in Assembly Systems,” Proceedings of the 4th International Conference on Changeable, Agile, Reconfigurable and Virtual Production (CARV2011), Montreal, Canada, October 2-5, 2011

[3] Y. Bozer and D. Ciemnoczolowski, “Performance evaluation of small-batch container delivery systems used in lean manufacturing – Part 1: system stability and distribution of container starts,” INT J PROD RES, vol. 51, pp. 555- 567, Jan. 2013.

[4] C. Marchwinski, “Following four steps to a lean material-handling system leads to a leap in performance,” Knowledge Center at http://www.lean.org

[5] H.S. Kilic, M.B. Durmusoglum, and M. Baskak, “Classification and modeling for in-plant milk-run distribution systems,” INT J ADV MANUF TECH, vol. 62, pp. 1135-1146, 2012.

[6] C. Harris and R. Harris, “Steps to Implementing a Lean Material Handling System,” March 2004, http://www.lean.org

[7] S.I. Satoglu and I.E. Sahin, “Design of a just-in-time periodic material supply system for the assembly lines and an application in electronics industry, INT J ADV MANUF TECH, vol. 65, pp. 319-332, 2013.

[8] R. Owens, “Advancing facility planning,” IND ENG, vol. 43, pp. 45-49, Nov. 2011.

[9] G.S. Brar and G. Saini, “Milk Run Logistics: literature review and directions,” Proceedings of the World Congress on Engineering, vol. I. London, U.K, July 6-8, 2011.

[10] M. Alnahhal and B. Noche, “Efficient material flow in mixed model assembly lines,” SpringerPlus, vol. 2, pp. 1-12, Aug 2013.

[11] M. Faccio, M. Gamberi, and A. Persona, “Kanban number optimization in a supermarket warehouse feeding a mixed-model assembly system,” INT J PROD RES, vol. 51, pp. 2997-3017, Feb 2013.

[12] M. Faccio, M. Gamberi, A. Persona, A. Regattieri, and F. Sgarbossa, “Design and simulation of assembly line feeding systems in the automotive sector using supermarket, kanbans and tow trains: a general framework,” J Manag Control, vol. 24, pp. 187–208, May 2013b.

[13] W. Sihn and K. Schmitz, "Extended Multi-Customer Supplier Parks in the Automotive Industry," CIRP ANN-MANUF TECHN, vol. 56, pp. 479– 482, Jun. 2007.

[14] Y. Wong and K. Wong, "Approaches and Practices of Lean Manufacturing: The Case of Electrical and Electronics Companies," AFR J BUS MANAG 5, pp. 2164- 2174, 2011

[15] G. Kasperk, J. Mangoldt, and R. Drauz, "Mixing Push & Pull to survive in China: Value Stream Design in an automotive case," Zukünftige Entwicklungen in der Mobilität, Gabler Verlag pp. 301-313, 2012.

[16] R. Owens, "Practical Facility Layout Technology for Industrial Engineers and Educators in Asia," Proceedings of the11th Asia Pacific Industrial Engineering and Management Systems Conference, Kuala Lumpur, Malaysia, December 7-10, 2010.

[17] N.H.A. Halim, A. Jaffar, N. Yusoff, and A.N. Adnan "Gravity Flow Rack's Material Handling System for Just-In-Time (JIT) Production," Procedia Engineering, vol. 41, pp. 1714 – 1720, 2012.

[18] R. Hanson and M.I. Johansson, "On the choice of approach for materials supply to mass customized line assembly," Proceedings of the PLANs conference, Jönköping, Sweden, pp. 97-110, September 5-6, 2007.

[19] E.J. Williams, O.M. Ulgen, S. Bailiff, and R. Lote, "Material handling resource utilization simulation study for stamping plant," Proceedings of the 38th conference on Winter simulation, California, USA, pp. 1940-1945, December 03- 06, 2006.

[20] M. Rother, J. Shook, J. Womack, and D. Jones, Learning to See Version 1.3. MA, Boston: Lean Enterprises Institution Inc 1999.

[21] J.P. Womack and D.T. Jones, Lean Thinking: Banish Waste and Create Wealth in Your Corporation. New York: Simon & Schuster, 2003.

[22] Y. Sugimori, K. Kusunoki, F. Cho, and S. Uchikawa, "Toyota production system and kanban system materialization of just-in-time and respect-for-human system," INT J PROD RES, vol. 15, pp. 553-564, 1977.

[23] R. Hanson and C. Finnsgård, "Impact of unit load size on in-plant materials supply efficiency," INT J PROD ECON, vol. 147, pp. 46-52, 2014.

[24] T. Rother, "Less empty runs with milk run and tugger train systems [Per Milkrun und Routenzugsystem Leerfahrten deutlich verringern] ," Maschinen Markt 51/52, 2011.

[25] C. Finnsgård, C. Wänström, L. Medbo, and W.P. Neumann, "Impact of materials exposure on assembly workstation performance," INT J PROD RES, vol. 49, pp. 7253-7274, Jan. 2011.

[26] W.P. Neumann and L. Medbo, "Ergonomic and technical aspects in the redesign of material supply systems: Big boxes vs. narrow bins" INT J PROD ECON 40, pp. 541-548, Aug. 2010.

[27] R. Hanson, "Effects of using minomi in in-plant materials supply," Journal of Manufacturing Technology Management, vol. 22, pp. 90 – 106, 2011

[28] E. Abele and F. Brungs, "Optimization of in-plant material provision via milk run [Optimierung der innerbetrieblichen Materialversorgung mittels Milkrun], " Industrie Management, vol. 25, pp. 61-64, Sep 2009.

[29] F. Alizon, Y. Dallery, I. Essafi, and D. Feillet, "Optimizing material handling costs in an assembly workshop," INT J PROD RES, vol. 47, pp. 3853-3866, May 2009.

[30] M. Michael and N. Claudia, "A Report on the Current Event on the WMS Market," WMS Market Overview. http://www.wmsfinder.com

[31] J. Golz, R. Gujjula, H.-O. Günther, S. Rinderer, and M. Ziegler, "Part feeding at high-variant mixed-model assembly lines," Flex Serv Manuf J, vol. 24, pp. 119–141, Jun. 2012.

[32] S. Emde and N. Boysen, "Optimally locating in-house logistics areas to facilitate JIT-supply of mixed-model assembly lines," INT J PROD ECON, vol. 135, pp. 393-402, Jul. 2012c.

[33] D. Battini, N. Boysen, and S. Emde, "Just-in-Time supermarkets for part supply in the automobile industry," J Manag Control, vol. 24, pp. 209-217, Jul. 2013.

[34] R. Domingo, R. Alvarez, M.M. Peña, and R. Calvo, "Materials flow improvement in a lean assembly line: a case study," ASSEMBLY AUTOM, vol. 27, pp. 141-147, 2007.

[35] R. Álvarez, R. Calvo, M.M. Peña, and R. Domingo, "Redesigning an assembly line through lean manufacturing tools," INT J ADV MANUF TECH, vol. 43, pp. 949-958, Aug. 2009.

[36] D. Ciemnoczolowski and Y. Bozer, "Performance evaluation of small-batch container delivery systems used in lean manufacturing – Part 2: number of Kanban and workstation starvation," INT J PROD RES, vol. 51, pp. 568-581, Jan 2013.

[37] B. Costa, L.S. Dias, J.A. Oliveira, and G. Pereira, "Simulation as a tool for planning a material delivery system to manufacturing lines," Proceedings of Engineering Management Conference IEEE International, Estoril, Portugal, June 28-30, 2008.

[38] S. Emde and N. Boysen, "Optimally routing and scheduling tow trains for JIT supply of mixed-model assembly lines," EUR J OPER RES, vol. 217, pp. 287-299, 2012b.

[39] S. Emde, M. Fliedner, and N. Boysen, "Optimally loading tow trains for JIT-supply of mixed-model assembly lines," IIE TRANS, vol. 44, pp. 121-135, Feb. 2012a.

[40] A. Kovács, "Optimizing the storage assignment in a warehouse served by milkrun logistics" INT J PROD ECON, vol. 133, pp. 312– 318, 2011.

[41] V. Ramesh, K.V. Sreenivasa Prasad, and T.R. Srinivas, "Study on implementation of one-piece lean line design using simulation techniques: A practical approach," J IND ENGINEERING, vol. 5, pp. 20-36, Jan. 2009.

[42] D. Battini, M. Faccio, A. Persona, and F. Sgarbossa, "Supermarket warehouses: stocking policies optimization in an assembly-to-order environment," INT J ADV MANUF TECH, vol. 50, pp. 775-788, Sep. 2010.

[43] T.A. Sargent, G. Michael, and M.G. Kay, "Implementation and utilization of a decentralized storage system: costing model," INT J OPER PROD MAN, vol. 15, pp. 210 – 219, 1995.

[44] S.I. Satoglu, B. Durmusoglu, and I. Dogan "Evaluation of the conversion from central storage to decentralized storages in cellular manufacturing environments using activity-based costing," INT J PROD ECON, vol. 103, pp. 616–632, Oct. 2006.

[45] V. Limère, H. Van Landeghem, M. Goetschalckx, E.-H. Aghezzaf, and L.F. McGinnis, "Optimizing part feeding in the automotive assembly industry: deciding between kitting and line stocking," INT J PROD RES, vol. 50, pp. 4046-4060, Mar. 2012b.